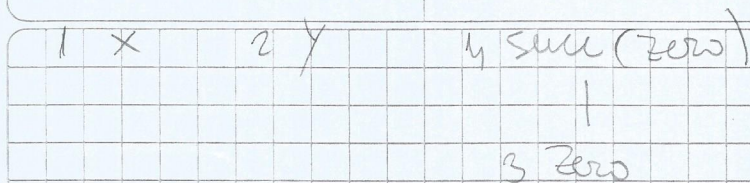


5 if x then y else $\text{succ}(\text{zero})$



1) $W(\text{zero}) \rightsquigarrow \emptyset \vdash \text{zero} : \text{Nat}$

2) $W(y) \rightsquigarrow y : y_1 \vdash y : y_1$

3) $W(x) \rightsquigarrow x_1 : X_1 \vdash x_1 : X_1$

4) $W(\text{succ}(\text{zero})) \rightsquigarrow \vdash \text{succ}(\text{zero}) : \text{Nat}$

$$\text{mgu}(\{\text{Nat} \equiv \text{Nat}\}) = \emptyset$$

5) $W(\text{if } x \text{ then } y \text{ else } \text{succ}(\text{zero})) \text{ (A)}$

$$\text{mgu}(\{\exists y_1 \equiv \text{Nat}, X_1 \equiv \text{Bool}\}) \xrightarrow{\text{elim } \exists y_1 \equiv \text{Nat} \Rightarrow}$$

$$\exists X_1 \equiv \text{Bool} \xrightarrow{\text{elim } \exists X_1 \equiv \text{Bool} \Rightarrow} \emptyset$$

$$\exists y_1 \equiv \text{Nat}, X_1 \equiv \text{Bool}$$

(A) $\rightsquigarrow X_1 : \text{Bool}, y_1 : \text{Nat} \vdash \text{if } x \text{ then } y \text{ else } \text{succ}(\text{zero}) : \text{Nat}$

6) $W(\lambda y. \text{if } x \text{ then } y \text{ else } \text{succ}(\text{zero})) \text{ (A)}$

(A) $\rightsquigarrow X_1 : \text{Bool} \vdash \lambda y : \text{Nat}. \text{if } x \text{ then } y \text{ else } \text{succ}(\text{zero}) : \text{Nat} \rightarrow \text{Nat}$

7) $W(\lambda x. \lambda y. \text{if } x \text{ then } y \text{ else } \text{succ}(\text{zero}))$

$\rightsquigarrow \vdash \lambda x : \text{Bool}. \lambda y : \text{Nat}. \text{if } x \text{ then } y \text{ else } \text{succ}(\text{zero}) : \text{Bool} \rightarrow (\text{Nat} \rightarrow \text{Nat})$